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Review

Concept, development and applications of DNA computation

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ABSTRACT

As traditional silicon-based chips approach their theoretical limits on computing power, DNA-based computation presents a promising alternative with potential advantages such as reduced size, high storage density, low consumption, long-term stability and the ability to perform in memory-computing. This review provides a summarized overview of DNA computation, covering its history, development and applications. In addition, we also highlight the possible directions to address the challenges that stand in the way of realizing the full potentials of this technology.

1. Introduction

DNA computation is an emerging computing paradigm that employs oligonucleotides instead of traditional silicon chips, while maintaining similar logical architectures to silicon-based computation. It has the ability to perform Boolean logic gates and numeric calculations, but with reduced size and increased gate throughput. As traditional silicon-based computers approach the physical limits of the miniaturization, the development of Moore's law is slowing down and may not be effective anymore [1,2]. In addition, the von Neumann bottleneck has demonstrated that increasing information exchange between the central processing unit and main memory would be another factor that restricts the full release of computation power [3].

Compared to traditional silicon-based systems that use Turing pattern and need data and commands to perform computation (Fig. 1a), DNA-based computation is similar to cellular automata whose states are simultaneously updated by a uniformly applied state-transition rule that refers to the states of their relative elements (neighbors). These features make DNA-based computation particularly well-suited for solving NP-complete tasks where only initial conditions of inputs with defined computing rules are needed (Fig. 1b). Although the limited speed at which a single computing operation occurs in DNA-based computation is primarily influenced by the hybridization and polymerization kinetics of the DNA strands, DNA-based systems have

enormous parallel computing potential with the ability to conduct trillions of operations simultaneously in small volume. Additionally, features of the information can be encoded in DNA sequence, allowing each input to contain more patterns than the simple 0–1 state. These distinctive features of DNA-based computation make it a promising tool for researchers and engineers to develop new computing systems.

Consequently, DNA computation has found its niche in the rapidly expanding market of big data processing and storage, including mathematics [4–10], electrical engineering [11–13], artificial intelligence [14–16], clinical diagnostics [17–21], and cryptography [22]. This mini-review will retrospectively examine the history of DNA computation (Fig. 1c), outline its longitudinal developmental stages, and summarize the diverse real-world applications as well as the perspective of DNA-based computation.

2. In the beginning: mathematical problems

A conventional computer is comprised of two primary functions: processing and storage. The processing module integrates and transforms information to generate new information, while the storage module is responsible for saving information [23]. Unlike the concept of separated implementation of processing and storage, DNA-based computing with no storage module was first introduced in 1994 when Leonard M. Adleman solved the Hamiltonian Path Problem (HPP) using multiple strands

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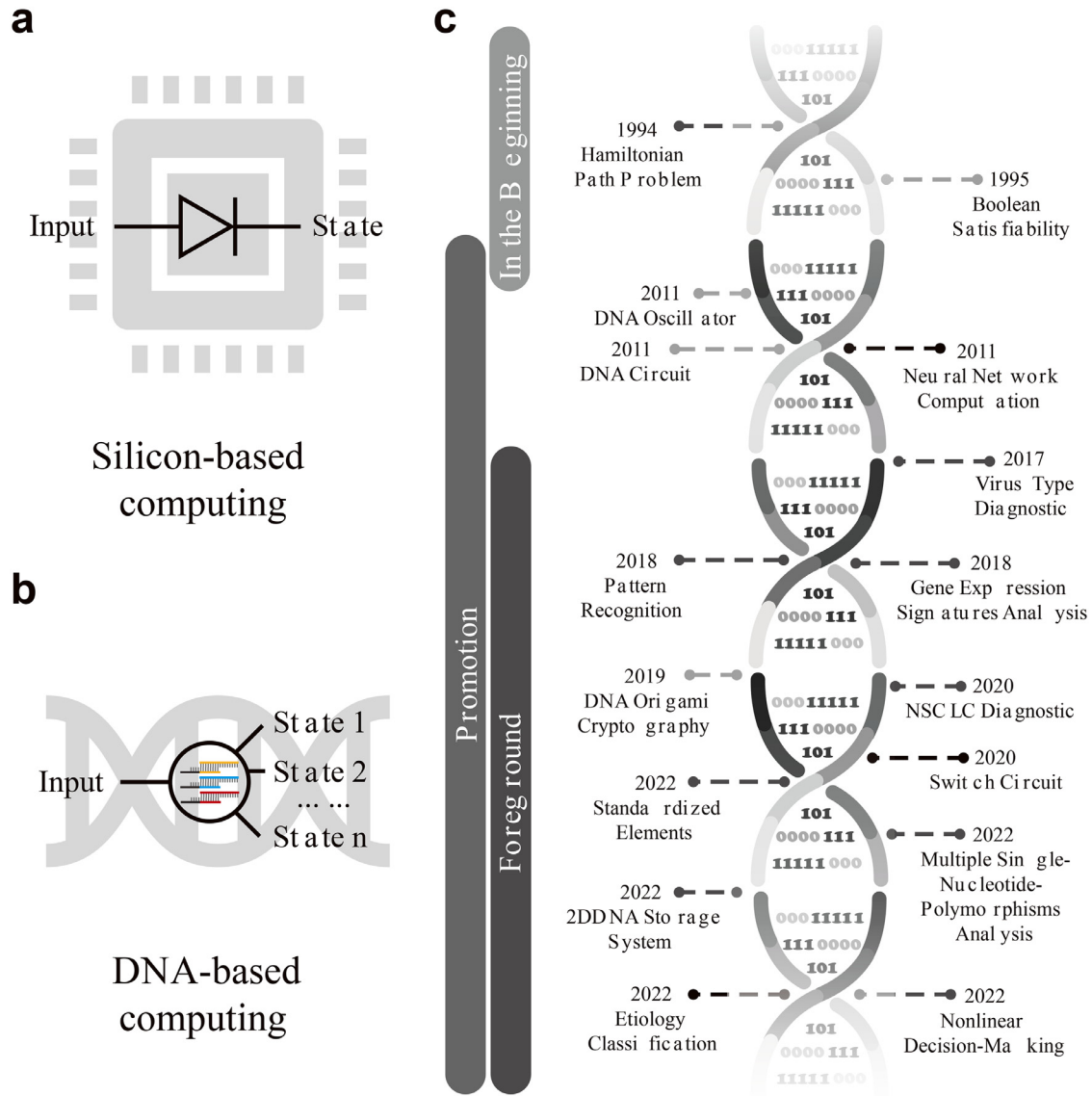


Fig. 1. (a) Silicon-based computing scheme and (b) DNA-based computing scheme. (c) The historic development of DNA-based computation.

of short DNA oligonucleotides via DNA ligation reaction and polymerase chain reaction (PCR) (Fig. 2), clearly demonstrating the reliability and efficiency of DNA computing [4]. Adleman created DNA molecules encoding possible Hamiltonian paths, and the answers were decoded via hybridized sequences and subsequently amplified by PCR using a specialized primer. He further suggested that the scalability potential and the number of operations per second during the ligation step would ultimately exceed the number of operations in supercomputers by thousand folds. In the same year, Lipton extended the computing strategy to satisfiability of a Boolean formula, opening a new era of DNA-based circuits [5]. This strategy standardized the computing unit into simple Boolean logic that is compatible with electronic engineering design patterns, greatly facilitating further transformation of DNA-based computing.

Soon after that, DNA computation displayed its potential for highly parallel searches which can be structured to exploit operations in numerous other mathematical problems such as encompassing three-coloring problems [6], quantified Boolean formulas [7], independent sets [8], knapsack [9], subgraph isomorphism and maximum clique [10]. Notwithstanding its limited operating speed, DNA computing has emerged as a feasible approach to tackling complex problems, particu-

larly those require significant amount of parallel calculations, such as NP-complete problems.

3. Promotion: engineering problems

As the volume of data being processed continues to grow in modern world, there is an increasing demand for factoring large numbers to achieve parallelism of massive computing and storage [22]. In addition, complex data analysis requires multi-modular collaboration. DNA computation has demonstrated its ability to handle heavy computational loads, leading researchers to explore its potential applications in various fields. Scientists have made effort to build an adder [11–13,24] or an oscillator [25–27] in order to verify the possibility to build a “computer” with DNA, and later they have managed to achieve complex computations using cascading DNA-based network [14–16].

One attempt to implement DNA-based computation involves designing digital Boolean logic gates with thresholding functions using multiple short DNA oligonucleotides [11]. The toehold-mediated strand displacement theory indicates that the reaction rate is dependent on the length of toehold, allowing DNA inputs only pass through the thresholding module first until sufficient amount over the threshold value to

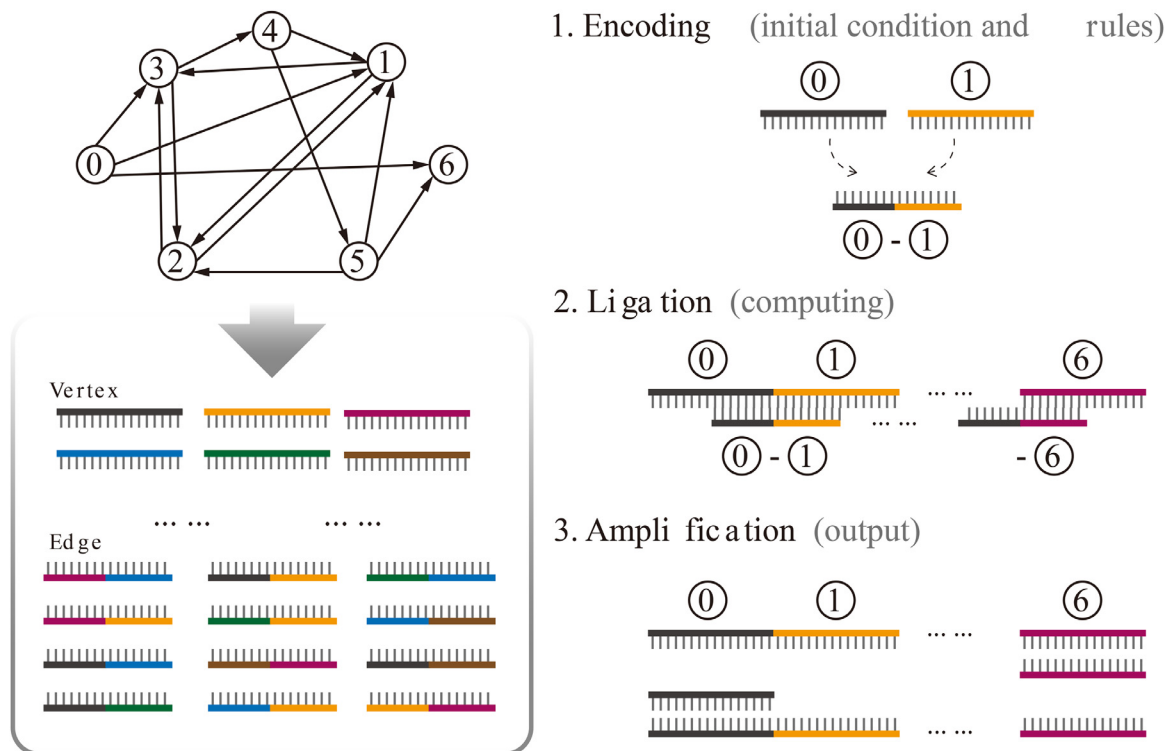


Fig. 2. Hamilton path problem solved by a DNA-base computation scheme [4].

trigger the output. This principle enables the implementation of “AND” logic with a threshold equal to twice the amount of two inputs and “OR” logic with a threshold equal to half the amount of two inputs (Fig. 3a). Additionally, various successful designs of full adders have been reported, such as enzyme-triggered sequence elongation or switch-mimicry strategy, which can limit the inter-stranded leakage to a narrow level. In specific, the enzyme-triggered strategy requires two converted signals to work together to implement “AND” and one converted signal with one fuel to implement “OR” (Fig. 3b) [24]. The switch circuit strategy, on the other hand, resembles an electronic circuit with “AND” logic implemented by two DNA switches in series (Fig. 3c) and “OR” logic by two switches in parallel [12].

Meanwhile, DNA-based oscillators offer various novel capabilities in DNA computing, such as generating precise timing signals to synchronize different operations and recognizing specific patterns, showing that DNA-based network can handle more complicated functions such as dynamic and continuous signals. Winfree’s group has designed synthetic oscillators using transcriptional regulatory elements such as promoters, repressors, and feedback loops (Fig. 3d). These oscillations could be tuned by adjusting the concentrations of regulatory elements, indicating their potential applications in synthetic biology and biotechnology [25]. Inspired by the silicon-based computer, organization of modular parts with different functions can generate more powerful functions for the entire system. A screening strategy for modular elements with standardized performance has recently been discussed, which helps employ modular parts to accurately predict and efficiently assemble complex functional circuits [27]. Soloveichik’s group developed an autonomous DNA oscillator that can switch between two states and a DNA-based keypad lock that requires the correct sequence of inputs (Fig. 3e). Since it is difficult to control enzyme behaviors in a cell, the enzyme-free approach is more scalable and easier to apply in a bio-system than that of the previously mentioned enzyme-based systems with its controllable preponderance and broad compatibility in reaction conditions. This work opened a new possibility for developing autonomous molecular machines that can perform complex tasks without enzymes [26].

Researchers have also explored DNA computation’s potential in artificial intelligence, such as pattern recognition and complicated decision-making. Neural network computation with DNA strand displacement cascades has been developed, leading to the creation of DNA-based winner-take-all (Fig. 3f) [14] and loser-take-all neural networks for applications such as pattern classification [15]. Recently, a nonlinear decision-making strategy has been introduced by Genot’s group, using DNA-encoded enzymatic neurons assembled in multilayer architectures to classify nonlinearly separable regions (Fig. 3g) [28]. The realization of these complex computations with DNA-based circuits shows that DNA computation has the possibility of being used for applications towards higher level artificial intelligence.

Unlike silicon-based computations in which all elements are driven by the same electric signal, DNA-based circuits can be operated by different inputs encoded in orthogonal oligonucleotide sequences. Therefore, Turing machine theory is suitable for traditional silicon-based computing pattern for its one by one searching while cellular automata is more likely to perform in DNA computation because it is easy to set up computing rules in each cellular. This allows for the computing in memory by eliminating the communication delay in traditional von Neumann architecture. Therefore, DNA-based computation shows considerable promise in addressing standard engineering and fundamental problems. In the immediate future, its potential ability to solve problems beyond the capabilities of current computers might position it as a highly promising emerging technology with immense possibilities.

4. Foreground: applications in a real-world scenario

As the further crossing of Information Technology (IT) and Biotechnology (BT), more attempts with DNA computation for real-world settings are emerging, such as DNA computing-based clinical diagnostics and cryptography which both need multiple recognition parameters and complex computation loads. DNA computation is well-suited for such tasks due to the ability to sense and analyze in one pot and has the potential to outperform the traditional analyze-post-sense methods. The

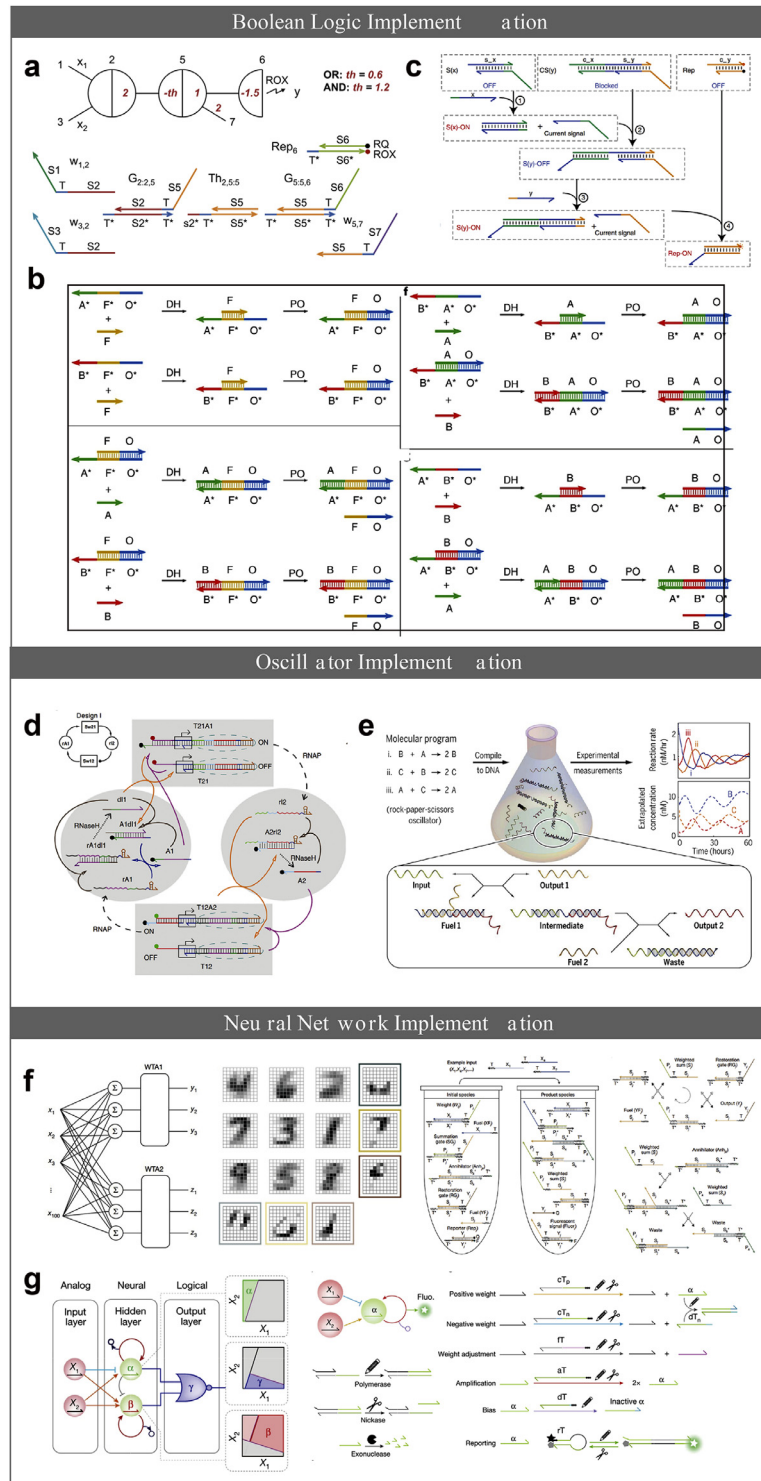


Fig. 3. Engineering problems solving with DNA circuits, included applications like calculator, oscillator and neural network tasks. (a) Boolean logic gates implement by threshold design [11]. Copyright 2011, AAAS Science. (b) Boolean logic gates implement by enzyme-triggered strategy [23]. Copyright 2018, Springer Nature. (c) Boolean logic implements via switching circuit [12]. Copyright 2020, Springer Nature. (d) Enzyme-triggered DNA oscillator [24]. (e) Enzyme-free DNA oscillator [25]. Copyright 2017, AAAS Science. (f) Pattern recognition based on DNA circuits [14]. Copyright 2018, Springer Nature. (g) Nonlinear decision-making with enzymatic networks [27]. Copyright 2022, Springer Nature.

integration of DNA computation with diagnostic processes into computing presents an opportunity to lower the technical barriers associated with disease diagnosis. When applied in point-of-care testing (POCT) for clinical diagnosis, this technology can streamline the diagnosis process, making it easier to generalize to the population.

In 2017, Macdonald's group made an attempt by using YES gate with fused complementary sequence to identify and report the diagnosis of rabies and other Lyssavirus types (Fig. 4a) [29]. Based on the idea of linear classification in machine learning, Seeling's group introduced a molecular computation strategy for analyzing complex gene expression

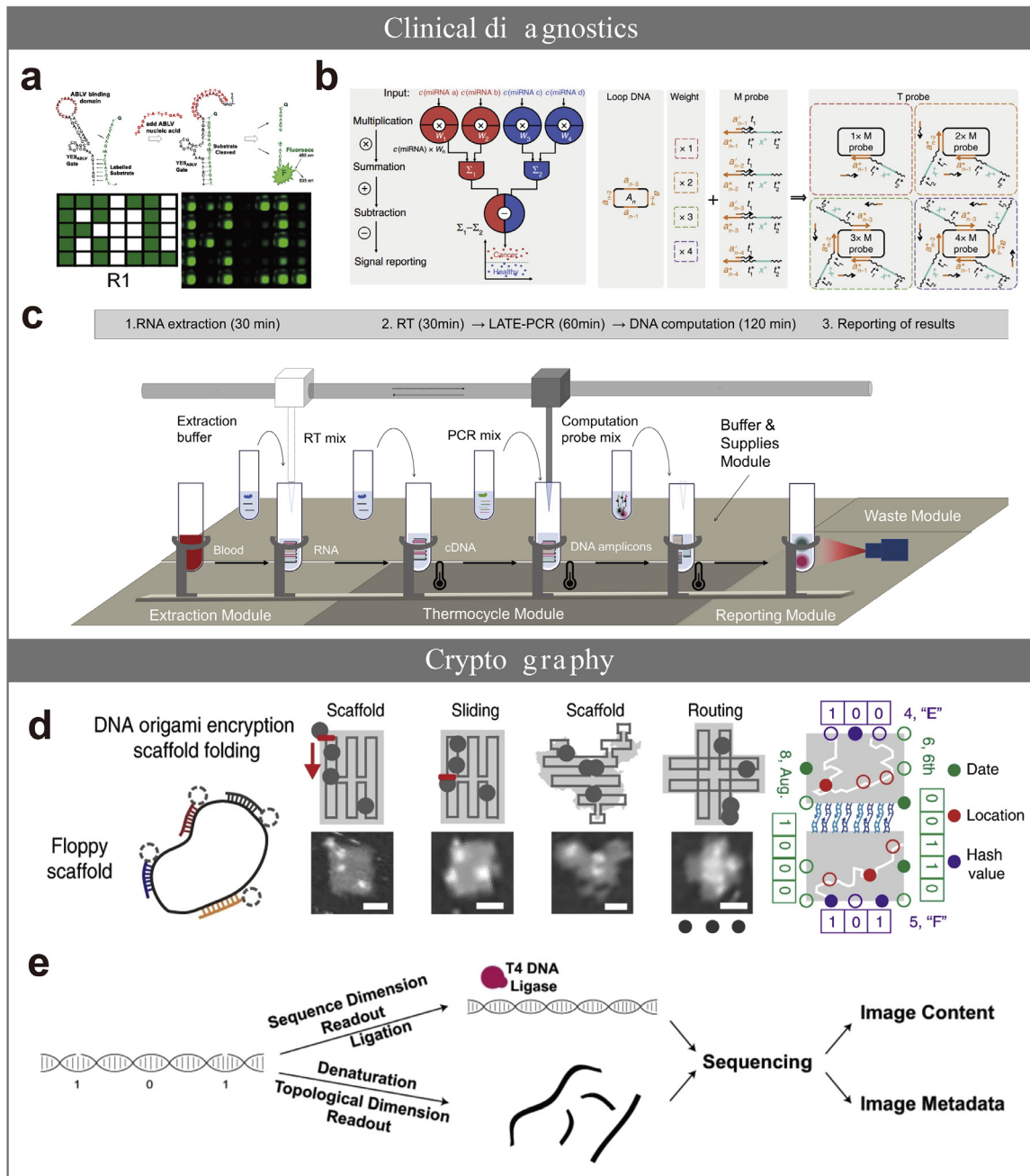


Fig. 4. Applications in clinical diagnostics and cryptography. (a) A prototype molecular automaton able to discriminate between seven different genotypes of Lyssaviruses, including Rabies virus [28]. Copyright 2017, Wiley-VCH. (b) Individual classification of healthy and non-small cell lung cancer (NSCLC) [18]. Copyright 2020, Springer Nature. (c) Etiology of acute respiratory illness classification [19]. (d) DNA origami cryptography (DOC) [29]. Copyright 2019, Springer Nature. (e) A DNA-based two-dimensional molecular data storage system (2DDNA storage system) [30]. Copyright 2022, Springer Nature.

signatures without requiring costly instrumentation [17]. Moreover, in 2020, Han's group demonstrated that DNA-based computing circuits, with the aid of in-silico training, could be used to classify healthy and non-small cell lung cancer in clinical samples (NSCLC) (Fig. 4b) [18]. Similarly, in 2022, they successfully classified the etiology of acute respiratory illness (Fig. 4c) [19] and analyzed complex single-nucleotide-polymorphism (SNP) combinations for rapid blood type determination [20]. These studies provide application scenarios for DNA computation in clinical diagnostics, revealing the potential for streamlined, intelligent, and convenient processes.

Likewise, cryptography, which involves encoding information to conceal messages, always requires extensive and economic computation power. In specific, additional storage space is typically required

for encryption for security reasons. Silicon-based information storage needs a lot of energy to maintain the stability of stored information. Therefore, high information density and low power consumption are important goals for cryptography. Due to the biostability and quaternary encoding of sequences, DNA computation is considered a suitable medium for encoding and decoding information. In addition, DNA possesses inherent structural information which can be combined with the above-mentioned sequence information to design and implement more complex crypto algorithms. In 2019, Fan's group developed a DNA origami cryptography (DOC) (Fig. 4d) which can create a key of over 700 bits. The transmitted message integrity can be ensured by establishing specific linkages between different DNA origamis carrying parts of the message [30]. In 2022, Schroeder's and Milenkovic's group reported

a DNA-based two-dimensional molecular data storage system (Fig. 4e) that recorded information in both the DNA sequences and backbone structures [31]. This 2D DNA storage system performs nontrivial joint data encoding, decoding, and processing by storing images in DNA sequences and embedding pertinent metadata as nicks in the backbone. Both examples demonstrate the ability of DNA computation to perform massive data storage and outstanding performance in cryptography.

Advancements in molecular-scale operations have led to the development of tasks such as data organization, access, manipulation, and other potential practical functionalities [32,33]. Therefore, DNA computation could be instrumental in shaping the future of information technology, particularly in the realms of processing and storing vast amounts of data.

5. Conclusions and outlook

As an emerging computing paradigm, DNA-based computation has been demonstrated to be an essential technology in the fields of computer science. Both the silico-based and DNA-based computing can be built by Boolean logic, which allows similar circuit designs in their standardization and modularization. To date, significant progress has been achieved in understanding the theories and practices of scaling up DNA circuits, which makes DNA computation more predictable and efficient.

Although DNA computation originally started with the demonstration of a computing application, it has shifted its goal away from being a competitor to electronic computers, toward several other promising research and application directions, especially on biomedical applications and information security. With respect to the former, researchers can program DNA modules to early detect or identify diseases with specific biomarkers, which can lead to more effective treatment options. With regard to the latter, DNA molecules possess unique properties that enable them to store vast amounts of information, and their inherent self-replication abilities make them highly resistant to damage or loss. Consequently, DNA-based encryption methods have emerged as highly secure options for data storage and transmission.

As a nascent technology, DNA computation is still facing various challenges, such as the relatively slow kinetics, liquid-based operation environment, stringent requirement on temperature and limited read-out strategies. Further research including engineered-enzyme for computing, regionalized organizing or surface reaction networks, imaging or sequencing technologies are required to comprehensively investigate the feasibility and efficacy of DNA computation, supported by strict theoretical and practical studies. With continued advancements, DNA computation has the potential to revolutionize the field of bio-engineering as well as information technology, leading to new and exciting opportunities for biomedical applications with big data processing and information storage.

Declaration of competing interest

The authors declare that they have no conflicts of interest in this work.

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Author profile

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